Class 23: Would It Replicate? — Instructor Notes

After Class 19 (confidence intervals) · one 75-min class · handout + skeleton notebook + slides · no data files

The idea. Three claims about p-values, all in one simulated creek whose truth students know. (1) A significant result does not predict a significant repeat: Team A's p = 0.03 replicates about half the time, and the original p makes no difference. (2) A significant result isn't necessarily real: how often "p < 0.05" comes from a creek with no excess depends on the number of sites and on how many creeks really have one. (3) A significant result isn't necessarily big: 20,000 logger readings per side make a 2 mg/L excess overwhelmingly significant.

What students produce. Committed predictions; a measured replication rate (overall and conditional on $0.01 \le p < 0.05$); one point on a class power curve; a whole-count table of true and false discoveries; CI/p counts for twenty surveys; a judgment of whether a significant excess is worth acting on, against a threshold they committed to in Part 1. **What this does not claim to teach:** power formulas, sample-size planning, Cohen's d, Bayesian replication probabilities, false-discovery-rate procedures, or anything about real Pennypack chloride. The world is simulated, and the handout and notebook both say so.

New material: the word "power," defined in plain terms (the fraction of repeated surveys that catch a real difference). The null distribution is built from the known creek SD rather than by shuffling. This is legitimate only because we built the world; with real data you shuffle (Class 15), and the notebook says so.

Materials. Handout, skeleton notebook, slide deck (Part 3 reveal, Part 5 recap, Part 7). The board needs a 4-column table for Part 4 (sites per side 6 / 12 / 24 / 48).

Source. Tidy Ecology, "What a p-value tells you, and what it does not" (28 Sept 2026, tidyecology.com/posts/what-a-p-value-tells-you). It uses a fenced/grazed seedling experiment with 12 plots per side and power ≈ 0.53. This activity keeps that arc with these changes: a chloride creek; a simulated null on the difference in means instead of the Welch t-test; percentile bootstrap CIs (Class 19) instead of t intervals; a class-built power curve; and a large-n segment the article does not have. Further reading: Goodman (1992), *Statistics in Medicine* 11(7):875–879; Cumming (2008), *Perspectives on Psychological Science* 3(4):286–300.

Timing (75 min firm)

Segment	Min	What happens
Part 1	4	Read Team A's report; Q1 individually, then compare; Q2; Q3 (commit a threshold). Stop bar. Collect a few Q1 numbers on the board.
Part 2	5	2.1–2.2 run; 2.3 a few times; loop at 10, Q4 prediction, 1000; log histogram.
Part 3	7	Pairs at 10 then 1000; Q5 prediction; fill the filter; Q6; cutoff; Q7. Replication slide.
Part 4	12	Assign sites per side; 4.1–4.2; Q8 prediction; 4.3 at 10 then 1000; teams post to the board; Q9.
Part 5	7	Q10 table from board numbers; Q11; recap slide.
Part 6	5	Run the prebuilt cell; Q12 and its follow-up.
Part 7	7	Three slides, a prediction before each; Q13.

Close	3	Q14 out loud; return to the board numbers from Q1.

That is 50 planned minutes against 75. Your class runs long, so plan on using the slack.

> **Watch out.** Never cut Part 3; it carries the answer to Q1. If short on time, cut in this order: Part 6 (show the twenty-survey plot on screen for one minute instead), then Q9 (fold it into the board discussion), then Part 5 becomes the recap slide only.

> **Note. Class 15 overlap.** Class 15 Part 3 showed p-values under no effect (about 5% significant) and a power ≈ 0.5 world (Q3.4: "what would you change?"). Part 2 here is a deliberate callback with a log axis. Part 4 answers Class 15′s Q3.4 by actually changing the number of sites.

The simulated creek and the numbers

Upstream mean 150 mg/L, downstream 183 mg/L (true difference 33), SD 40 at every site, 12 sites per side unless Part 4 changes it. Statistic: downstream mean − upstream mean, two-sided (`np.abs`, as in Class 15). All numbers below were run in ds313 (Python 3.13.11, datascience 0.18.1, NumPy 2.4.1): the filled notebook at every sites value, plus vectorized checks with 200,000–400,000 simulated surveys. Scripts and logs are in `_scratch/`.

Standard error of the difference	$40\sqrt{2/12} = 16.33$; true difference is 2.02 SE
Null distribution (10,000)	SD 16.1–16.6; the 95th percentile of |diff| (cell 3.3) is about **32** (31.3–32.7 across runs)
Power, $P(p < 0.05)$	**0.52** (0.524 in 400,000 surveys; one team's 1000 surveys: 0.50–0.55)
p-value percentiles (10th / 50th / 90th)	0.0009 / 0.043 / 0.45 (article: 0.0016 / 0.042 / 0.39)
Pairs kept (0.01 ≤ p < 0.05) of 1000	median 236 (5th–95th: 207–263)
Their replicates with p < 0.05	**≈ 0.52**: 0.47–0.57 for 90% of team runs. Large simulation: flat at 0.52 in every original-p band
Team A's 35 mg/L	gives p ≈ 0.03 against this null (34 → 0.037, 35 → 0.032, 36 → 0.027)
Notebook run time	Each null build (10,000 surveys) 0.1 s locally; every 1000-survey loop under 1 s. Hub not yet timed.

> **Note. Why one half, and why conditioning does nothing.** A survey is significant when its difference exceeds about 32 mg/L. Differences are centred on the truth, 33, so just over half land above 32. A replicate is a fresh draw from the same creek, so the original's p-value cannot change its chances. That is the whole lesson of Part 3, and Q7 asks students to say it.

Note. Goodman's number is not the answer, even though it looks close. Goodman (1992) assumes the true effect equals the observed one. Under that assumption a replicate of a p = 0.03 result reaches p < 0.05 with probability **0.59** (p = 0.05 → 0.50, p = 0.01 → 0.73). That is *observed power*, a function of p alone. The replicate's real chance is the power at the **true** effect, which p cannot reveal. The two agree here only by coincidence (p near 0.05, true power near 0.5). Pooling 200,000 originals by p band:

original p	< 0.001	0.001–0.01	0.01–0.05	0.05–0.2	> 0.2
replicates p < 0.05	0.519	0.526	0.524	0.528	0.527
Goodman (avg.)	0.96	0.82	0.62	0.38	0.14

If a student brings up "observed power," this table is the answer.

Part 1. The Report (≈ 4 min)

1 Any committed number. Expect 95–97% (reading p as "the chance it was a fluke"), with some guesses around 70–80%. Write the spread on the board; you come back to it at the close.

2 Common answers are "1 − p" and "it was significant, so it's real." Both are fine to commit to. Don't correct them yet.

3 Any number with a reason. There is no right answer. Insist that every student writes one before the notebook opens; Part 7 depends on it. Note the class's smallest value for later.

Part 2. Run the Creek (≈ 5 min)

4 Actual fraction about 0.50–0.55. The smallest p is usually 0 (shown at 0.0001) and the largest is above 0.98. The p-values span four orders of magnitude for one unchanging creek. *What changed:* which 24 sites got sampled, so the means, so the difference, so the p.

Note. The log histogram uses half-decade bins. With a 10,000-survey null, every p-value is a multiple of 0.0001. The red 0.05 line falls inside the bin from 0.032 to 0.1, so students should read the fraction from the printed output, not from the bars.

Part 3. Only the Surveys Like Team A's (≈ 7 min)

5 Most teams circle HIGHER ("those were already significant"). The answer is ABOUT THE SAME.

6 Kept about 236 of 1000. Their replicates: about 0.52 (one team may see 0.47 or 0.57). All replicates: about 0.52. The two fractions differ only by noise.

7 Cutoff ≈ 32 mg/L. (a) Observed differences scatter around the true 33, and 33 sits just above the cutoff, so a bit more than half of surveys clear it. (b) The replicate draws new sites. It "doesn't know" the original's p, so its chance of clearing 32 is the same as any survey's.

Watch out. If a team leaves the filter as `are.between(..., ...)`, the cell errors, which is fine. If they type `are.between(0.01, 0.5)`, the conditional fraction is still about 0.52, which accidentally makes the point. Check that they typed 0.05.

Slide: replication. Repeat rate by original-p band (the table above, without the Goodman row). Ask for a show of hands on Q5 before revealing it.

Part 4. More Sites

(≈ 12 min)

Assign one sites-per-side value to each team: **6, 12, 24, 48**. The 12-site team reproduces Part 2 with a fresh null, which is a useful check on the others. Name the real-creek fraction "power" once teams post it.

8 Verified values (one team's 1000 surveys falls in the range shown 90% of the time):

sites per side	6	12	24	48
cutoff (mg/L)	45.4	31.9	22.6	16.0
real creek: $p < 0.05$	0.30 (0.27–0.33)	0.53 (0.50–0.55)	0.82 (0.79–0.84)	0.98 (0.97–0.99)
no-excess creek: $p < 0.05$	0.050 (0.04–0.06)	0.051 (0.04–0.06)	0.050 (0.04–0.06)	0.050 (0.04–0.06)

Full notebook runs gave real/no-excess fractions of 0.273/0.056 (6), 0.497/0.064 (12), 0.809/0.039 (24), and 0.981/0.045 (48).

9 The real-creek row climbs from about 0.3 to nearly 1; the no-excess row stays near 0.05. More sites narrow the null, so the cutoff drops (45 → 16 mg/L). The true 33 doesn't move, so it clears a lower bar more often. A no-excess creek's differences shrink by the same factor as the cutoff, so 5% of them still clear it: the cutoff *is* the 95th percentile of that creek's surveys.

Note. If a student asks why not always use 48 sites: cost. Every site is a sample, a trip and a lab analysis. Choosing the number of sites before going out is what "power" is used for in practice. This activity does not teach the calculation.

Part 5. How Many Discoveries Are False?

(≈ 7 min)

10 Using the verified powers (students' board numbers will differ slightly):

	6 sites	48 sites
real creeks with $p < 0.05$	30	98
no-excess creeks with $p < 0.05$	45	45
total with $p < 0.05$	75	143
share with no real excess	**60%**	**31%**

11 No. The 5% is the false-alarm rate *among no-excess creeks*. Among significant results, 60% (6 sites) or 31% (48 sites) came from no-excess creeks. Few sites makes it worse because real creeks rarely reach significance. The other lever is how common real excesses are: if half the creeks had one, the false share would be about 14% (6 sites) or 5% (48 sites). Even perfect power leaves about 31% false when only 1 in 10 creeks has a real excess.

Note. In the article's terms the 100-of-1000 is the "prior." Students don't need the word. The 48-site column is the surprise: power alone can't fix a low base rate. Large-simulation check (12 sites): 46.4% false at 10% real, 22.4% at 25%, 8.8% at 50%.

Slide: recap. The two columns as a stacked bar (real hits vs false alarms), 6 vs 48 sites.

Part 6. Twenty Surveys, Twenty Intervals (≈ 3 min)

The prebuilt cell uses `np.random.seed(120)`. The twenty surveys reproduce exactly in ds313 (NumPy 2.4.1) as in the original draft environment (NumPy 1.26.3). The p-values come from each team's own null, so they differ in about the third decimal place. Values below use a 200,000-survey null.

#	diff	95% CI	p	#	diff	95% CI	p
1	25.9	−9.2, 62.2	0.112	11	35.3	11.0, 59.8	**0.030**
2	49.0	19.8, 77.7	**0.003**	12	62.6	27.4, 98.4	**0.0001**
3	29.4	−5.6, 67.7	0.072	13	34.2	6.8, 64.0	**0.036**
4	4.7	−17.1, 26.1	0.774	14	40.9	10.4, 68.9	**0.012**
5	15.7	−18.4, 45.7	0.338	15	20.2	−9.3, 48.2	0.217
6	46.5	12.6, 82.1	**0.004**	16	44.6	13.8, 73.5	**0.006**
7	44.3	17.1, 70.8	**0.007**	17	37.3	7.7, 68.4	**0.022**
8	22.3	−6.3, 53.0	0.174	18	37.6	3.3, 66.4	**0.021**
9	4.3	−28.1, 36.2	0.791	19	24.7	−3.3, 52.9	0.130
10	39.3	9.4, 70.4	**0.016**	20	27.7	1.7, 54.0	0.089

> **12** Contain 33: **19** (survey 4 misses). $p < 0.05$: **11**. Entirely above 0: **12**. The disagreement is **survey 20**: its interval [1.7, 54.0] excludes 0, but $p \approx 0.09$.
>
> *Follow-up:* survey 20′s difference, 27.7 mg/L, is below the 32 mg/L cutoff, so the test calls it not significant. The test's cutoff comes from the creek's true SD and is the same for every survey. The interval sees only survey 20′s 24 sites, which happened to be less spread out than the creek (sample SDs 30.6 upstream and 37.0 downstream, against a true 40). So its half-width is only about 26 mg/L, and the interval just clears 0. Accept any answer that names the two different sources of spread: the model's true SD versus this sample's own spread.

Watch out. Survey 20, and why CIs and p disagree here. With n = 12 the percentile bootstrap runs narrow. Over 4,000 simulated surveys its 95% intervals covered 33 only **92.1%** of the time; median width 60.0 mg/L against 64.0 from normal theory with the known SD. Intervals exclude 0 in 57.9% of surveys while $p < 0.05$ occurs in 53.7%, and the two calls disagree in **9.3%**. So "interval excludes 0 ⇔ $p < 0.05$" is only approximate here; don't present it as exact.

If a student asks whether a permutation test (Class 15) would fix this: no. A permutation p-value paired with the same bootstrap intervals still disagreed in **8.9%** of 3,000 surveys (10.9% for the known-SD null in the same run). The test spreads under "no excess" by pooling both sides; the interval resamples each side around the observed difference. They answer different questions, so they disagree near the cutoff (`_scratch/verify_perm_vs_boot.py`).

Note. Can students use a CI instead of a permutation test? Approximately: "0 outside the 95% bootstrap interval ⟹ the means differ at about the 5% level." Inferential Thinking presents this method, and the interval also gives the size of the difference. The approximation is poor with small samples. In this creek (SD 40), with **no** real difference, the CI rule raised a false alarm in 11.2% of surveys at 6 sites per side, 8.0% at 12, 6.2% at 24 and 6.6% at 48 (4,000 surveys each). The permutation test stayed at

5.0–5.9%. With a true difference of 2 standard errors at every size, the two verdicts disagreed in 17% of surveys at 6 sites, 8.5% at 12, 4.8% at 24 and 3.7% at 48 (2,000 each), nearly always in the direction "interval says different, test doesn't." The cause is the percentile bootstrap: a small sample understates its population's spread, so the interval runs narrow. Survey 20 is that case. The textbook's duality is still correct: "reject if x is outside the 95% CI" is exact by construction, and it rejects a true null 5% of the time whenever the interval really covers 95%; the percentile bootstrap only approximately does. A bias-corrected (BCa) interval does not fix this, because these data have no bias or skew to correct (11.4% false alarms at 6 sites, 7.0% at 12), whereas a bootstrap-t or Welch t interval accounts for the uncertain spread and holds about 5% at every size (`_scratch/verify_ci_methods.log`). Guidance for students: with roughly 20 or more per group, testing with the interval is fine and more informative; with fewer, use the permutation test for the yes/no question and read the interval as a somewhat-too-narrow estimate of size. Either way, a borderline result is reported as borderline. Scripts: `_scratch/verify_perm_vs_boot_byN.log`, `_scratch/verify_ci_false_alarm.log`.

Part 7. A Very Big Survey (slides) (≈ 7 min)

Same SD (40 mg/L); true excess **2 mg/L** (0.05 SD); 20,000 readings per side. Figures are in the `fig_B*.png` files and come from `_scratch/partB_figures.py`. Show B1, take the Q13(a) vote; show B2, take (b); show B3, have students copy the interval and compare it with their Q3 number.

Observed difference	2.01 mg/L (0.050 SD)
Null (10,000 surveys)	SD 0.40 mg/L; largest |difference| 1.52 mg/L
p-value	**p < 0.0001**: none of the 10,000 null surveys was as extreme (normal theory z = 5.0)
95% bootstrap CI	**1.25 to 2.78 mg/L** (the figure label rounds to 1.3 to 2.8)
How typical	This design gives p < 0.05 in 99.9% of surveys and p < 0.001 in 97%

13 (a) Most say NO or CAN'T TELL; the histograms are visually identical. (b) TINY. (c) 1.3 to 2.8 mg/L. Unless someone committed to a threshold under about 3 mg/L in Q3, the excess is real and not worth acting on. "Significant" answers *is it zero?*, not *is it big?*

Watch out. Disclose the seed if asked. Seed 1905 was chosen from 60 candidates so the observed difference (2.01) sits near the truth; a random seed would also give a tiny p almost every time. **Logger caveat:** real logger readings are autocorrelated, so 20,000 readings carry far less information than 20,000 independent sites. Treating them as independent overstates certainty. The simulation draws them independently; say so if a student who knows loggers objects.

Close (≈ 3 min)

14 Example: "Our p = 0.03 says a difference this big would be unusual if there were no downstream excess. It doesn't give you a 97% chance of agreeing: your chance is set by the true excess and your number of sites, and our p-value can't tell you either. It also doesn't say how big the excess is. For that, look at our interval." Push back on "assume the truth equals our 35 mg/L": that is Goodman's observed power, which only restates p.

> **Note.** Go back to the Q1 numbers on the board. Ask what 1 − p actually is. It is the chance that a **no-difference** creek gives a smaller difference than Team A's, which says nothing about Team B.

If You Finish Early

> **A** About **45** mg/L (45.2 in a full notebook run; 45.4 in 400,000 surveys), against the true 33. About 64% of the significant surveys report 40 or more; non-significant surveys average about 19 mg/L. Printing only $p < 0.05$ inflates published effects (the "winner's curse").

Appendix: solution code (instructor only)

```
# 2.4, 3.1, 4.3: change np.arange(10) to np.arange(1000)

# 3.2
like_team_a = pairs.where('original p', are.between(0.01, 0.05))

# 4.1
sites = 6    # or 12, 24, 48 as assigned
```